Drone Simulation Activity

Activity: Explore the principles of drone autonomous systems and their configuration using virtual simulation tools, even if a physical drone is not available.

Materials:

- Computer with drone simulation software (e.g., Gazebo, AirSim)
- Virtual environment with obstacles (e.g., 3D models of buildings, trees)

Instructions:

1. **Familiarise with Drone Simulation Software:**
   - Learn how to use the simulation software to create virtual drones and environments.
   - Understand the basic concepts of drone flight dynamics, sensors, and autonomous controls.
2. **Configure Virtual Drone:**
   - Set up the virtual drone with desired specifications (e.g., size, weight, sensors).
   - Configure autonomous flight modes and obstacle avoidance parameters.
3. **Create Virtual Environment:**
   - Design a virtual environment with various obstacles (e.g., buildings, trees, other drones).
   - Adjust the environment's complexity based on your learning objectives.

4. **Test Autonomous Flight:**

- Simulate the drone's flight in the virtual environment.
- Observe how the drone responds to obstacles and its ability to maintain a stable flight path.
- Experiment with different configurations to optimise performance.

5. **Data Collection and Analysis:**

- Record the simulation data, including sensor readings, altitude and flight path.
- Analyse the data to assess the drone's autonomous capabilities and identify areas for improvement.

6. **Final Project:**

- Create a comprehensive report or presentation summarising your findings from the simulation.
- Include the following:
    - A description of the virtual drone and environment
    - The challenges and successes encountered during testing
    - Data analysis and results
    - Recommendations for improvements or future research